HOUSE PRICE PREDICTION USING MACHINE LEARNING

Comprehensive Technical Project Report

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Independent Machine Learning Project
Dataset Source: Kaggle - House Prices: Advanced Regression Techniques
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Declaration

I hereby declare that this report titled 'House Price Prediction Using Machine Learning' is an independent project carried out by me for learning and portfolio development purposes. The work presented in this report is based on my implementation, experimentation, analysis, model development, evaluation and interpretation using the Ames Housing Dataset from Kaggle.

Abstract

This project aimed to develop an end-to-end machine learning pipeline capable of accurately predicting residential property prices. The project was designed not only to achieve high predictive performance but also to develop a deeper understanding of data preprocessing, feature engineering, model selection, validation methodology, and explainable AI. Multiple machine learning models were evaluated, including Linear Regression, Ridge Regression, Lasso Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost. A tuned XGBoost model emerged as the final model with a cross-validated R^2 score of 0.9057. SHAP explainability was used to understand model behavior and identify the major factors influencing house prices.

1. Introduction

House price prediction is one of the most widely studied regression problems in machine learning. Real estate markets are influenced by numerous variables including property size, construction quality, age, neighborhood, amenities, garage capacity, renovation history and economic factors. Predicting property prices accurately is valuable for buyers, sellers, investors, financial institutions and real-estate platforms. This project was undertaken to build a complete machine learning workflow that mirrors real-world data science practice. Rather than focusing solely on predictive accuracy, emphasis was placed on data understanding, feature engineering, validation, explainability and business interpretation.

2. Problem Statement

The objective of this project was to develop a machine learning model capable of estimating residential property prices using structured housing attributes. The model should generalize well to unseen properties while providing interpretable insights regarding the factors that influence housing prices.

3. Dataset Description

The dataset used in this project was obtained from the Kaggle competition 'House Prices: Advanced Regression Techniques'. The competition dataset is based on the Ames Housing Dataset and contains detailed information about residential properties. The dataset consists of 1460 observations and 81 variables, including the target variable SalePrice. Features cover property dimensions, quality ratings, basement information, garage characteristics, neighborhood information, building materials, year of construction and renovation history. The richness of the dataset makes it ideal for demonstrating advanced machine learning workflows.

4. Project Objectives

1. Understand the dataset structure and quality.
2. Perform comprehensive exploratory data analysis.
3. Engineer meaningful features from raw attributes.
4. Build preprocessing pipelines to handle mixed data types.
5. Compare multiple machine learning algorithms.
6. Optimize model performance through hyperparameter tuning.
7. Prevent data leakage through proper validation strategies.
8. Explain model predictions using SHAP.
9. Derive business insights from model outputs.
10. Develop an interview-ready machine learning project.

5. Data Understanding

The first stage involved examining the dataset using shape, head, info and descriptive statistics. Numerical and categorical variables were identified separately. Understanding the structure of the dataset helped determine preprocessing requirements and guided subsequent analysis. During this stage, attention was given to feature types, missing values, outliers and target variable characteristics.

6. Exploratory Data Analysis

EDA was conducted to uncover patterns, relationships and anomalies. The SalePrice distribution was analyzed using histograms and boxplots. Correlation analysis was used to identify variables strongly associated with SalePrice. Missing value visualizations were studied to understand data quality issues. Scatter plots and box plots were used to identify influential variables and potential outliers. EDA served as the foundation for feature engineering and model selection decisions.

7. SalePrice Distribution Analysis

The target variable SalePrice exhibited a strong positive skew. The mean value exceeded the median value, indicating the presence of expensive outlier properties. The skewness value was approximately 1.88. Such skewness can negatively affect regression performance because many models assume approximately normal target distributions. Histograms and boxplots confirmed a long right tail. This observation motivated the use of logarithmic transformation during preprocessing.

8. Correlation Analysis

A correlation matrix was generated to identify relationships between numerical variables and SalePrice. OverallQual, GrLivArea, GarageCars, GarageArea, TotalBsmtSF and YearBuilt emerged as some of the strongest predictors. These findings were consistent with real-world housing economics, where quality, size and property age significantly influence value.

9. Missing Value Analysis

A substantial number of variables contained missing values. However, domain investigation revealed that many missing values represented the absence of a feature rather than incomplete data. For example, missing PoolQC values typically indicated that a property did not contain a pool. Similarly, missing garage attributes often indicated the absence of a garage. Recognizing this distinction was important because naive deletion could remove useful information.

10. Feature Engineering

Feature engineering was performed to create higher-level variables capable of capturing more meaningful relationships. TotalSF combined basement and floor areas to represent overall living space. HouseAge measured the age of the property at the time of sale. RemodelAge quantified the time since renovation. TotalBath combined full and half bathrooms into a single metric. HasGarage captured whether a property possessed a garage. These engineered features provided richer information than the original attributes alone.

11. Target Transformation

Because SalePrice was highly skewed, a logarithmic transformation using log1p(SalePrice) was applied. The transformation reduced skewness, stabilized variance and improved compatibility with regression models. Log transformation is a common practice in housing price prediction because property values often exhibit exponential rather than linear behavior.

12. Data Preprocessing Strategy

The dataset contained both numerical and categorical variables, requiring different preprocessing strategies. Numerical variables were processed using median imputation and RobustScaler. Median imputation was chosen because it is resistant to outliers. RobustScaler was preferred over StandardScaler because housing datasets contain many extreme observations. Categorical variables were processed using most-frequent imputation and one-hot encoding. All preprocessing steps were organized within Scikit-Learn Pipelines and ColumnTransformers.

13. Pipeline Architecture

A major objective of the project was to implement a production-style preprocessing workflow. Pipelines ensured that preprocessing transformations were applied consistently during training and evaluation. ColumnTransformer allowed different preprocessing operations to be applied to numerical and categorical variables. This architecture improved reproducibility and reduced the risk of accidental data leakage.

14. Model Building

Several machine learning algorithms were evaluated. Linear Regression served as the baseline model. Ridge and Lasso Regression were used to evaluate regularized linear models. Decision Trees were explored because of their interpretability. Random Forest and Gradient Boosting were investigated as ensemble methods. XGBoost was selected because of its strong performance on structured tabular datasets.

15. Model Comparison

The following R^2 scores were obtained on the train-test evaluation: Linear Regression 0.9129, Ridge Regression 0.9083, Lasso Regression 0.8990, Decision Tree 0.8131, Random Forest 0.8874, Gradient Boosting 0.8995, XGBoost 0.9045 and Tuned XGBoost 0.9094. These results indicated that multiple models achieved competitive performance, making robust validation essential.

16. Overfitting Investigation

Random Forest achieved a training score of approximately 0.981 and a testing score of approximately 0.887. The large gap indicated overfitting. This observation highlighted an important machine learning principle: excellent training performance does not guarantee strong generalization. Understanding this distinction influenced later model selection decisions.

17. Hyperparameter Tuning

RandomizedSearchCV was used to optimize XGBoost. Parameters such as n_estimators, max_depth, learning_rate, subsample and colsample_bytree were explored. The best configuration included subsample=0.8, n_estimators=500, max_depth=4, learning_rate=0.05 and colsample_bytree=0.7. Hyperparameter tuning improved stability and predictive performance.

18. Discovery of Data Leakage

One of the most important lessons from this project was the identification and correction of data leakage. Initially, preprocessing transformations were applied before cross-validation. This allowed information from validation folds to influence preprocessing statistics. The methodology was corrected by embedding preprocessing inside Scikit-Learn Pipelines before performing cross-validation. This ensured that validation data remained completely unseen during training.

19. Cross Validation

Proper 5-fold cross-validation was performed after correcting the leakage issue. The results were: Linear Regression 0.8421, Ridge Regression 0.8634 and Tuned XGBoost 0.9057. These results revealed that Tuned XGBoost generalized substantially better than the linear models when evaluated robustly.

20. Final Model Selection

Although Linear Regression achieved a strong train-test score, cross-validation demonstrated that Tuned XGBoost provided the best balance between predictive accuracy and generalization. Therefore, Tuned XGBoost was selected as the final model.

21. Feature Importance Analysis

Feature importance scores were extracted from XGBoost. OverallQual emerged as the most influential variable. Other important variables included GarageFinish_Unf, GarageCars, GrLivArea, KitchenQual and CentralAir. These results indicated that quality and amenity-related characteristics strongly influence residential property values.

22. SHAP Explainability

SHAP (SHapley Additive Explanations) was used to provide model interpretability. Unlike traditional feature importance methods, SHAP explains both the magnitude and direction of feature influence. The transformed dataset contained 287 encoded and engineered features. SHAP values were computed to identify which features contributed most strongly to predictions.

23. SHAP Findings

OverallQual emerged as the dominant feature. GrLivArea, OverallCond, TotalBsmtSF, YearRemodAdd and neighborhood variables also played major roles. An interesting finding was that renovation history appeared more influential than original construction year. This suggests that updated homes can compete effectively with newer properties. The SHAP results aligned closely with real-world housing economics, increasing confidence in the model.

24. Business Insights

The project revealed several practical insights. Construction quality consistently emerged as the strongest determinant of house prices. Larger living areas and basement areas significantly increased property value. Neighborhood characteristics remained highly influential. Renovation history was more important than expected, indicating that improvements can substantially affect market value. Garage capacity and overall property condition also contributed meaningfully to pricing decisions.

25. Challenges Faced

Several challenges were encountered throughout the project. These included handling extensive missing values, preprocessing mixed feature types, controlling overfitting, preventing data leakage, selecting reliable evaluation methodologies and balancing predictive performance with interpretability. Addressing these challenges improved both the technical quality of the project and understanding of machine learning workflows.

26. Key Learnings

This project demonstrated that machine learning success depends on much more than algorithm selection. Feature engineering, preprocessing quality, validation methodology and interpretability often contribute more than model complexity alone. The project reinforced the importance of cross-validation, pipeline design, leakage prevention and explainable AI.

27. Future Improvements

Potential future enhancements include model serialization using Joblib, development of a Streamlit web application, cloud deployment, automated retraining workflows, interactive dashboards and continuous model monitoring. Additional experimentation with LightGBM and stacking ensembles may further improve performance.

28. Conclusion

This project successfully developed an end-to-end machine learning pipeline for residential property valuation. Through systematic data analysis, feature engineering, preprocessing, model comparison, hyperparameter tuning, validation and explainability, a tuned XGBoost model achieved a cross-validated R^2 score of 0.9057. The project not only produced a high-performing predictive model but also provided valuable insights into the factors that influence housing prices. The combination of strong predictive performance, proper validation methodology and explainability makes this project a comprehensive demonstration of practical machine learning skills.

Appendix: Interview Questions

Why was log transformation used? Why choose RobustScaler over StandardScaler? What is data leakage and how was it fixed? Why was XGBoost selected? What is cross-validation? How does SHAP differ from feature importance? Why can a model with a higher train-test score still be inferior? These are key interview questions that can be answered directly using this project.

References

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